SDS188 A1S2 - EXCEL FORMULAS ONLY - AUDITED FINAL

Chapters 6-8 full core | Ch9 formal: Slides 1-51 / Lesson 9.7 + official Practice Q4 power bridge only

C=confidence; alpha=1-C; n=sample size; N=population size; e=margin; p=sample proportion; pi=population proportion; df=n-1. ALL cumulative DIST functions below return LEFT area.

CORE HELPERS

n
```
=COUNT(range)
```

Xbar
```
=AVERAGE(range)
```

sample s
```
=STDEV.S(range)
```

square root
```
=SQRT(value)
```

required n: round UP
```
=ROUNDUP(value,0)
```

absolute value
```
=ABS(value)
```

CHAPTER 6 - CONTINUOUS DISTRIBUTIONS

NORMAL / STANDARD NORMAL

Normal LEFT P(X<x)
```
=NORM.DIST(x,mu,sigma,TRUE)
```

Normal RIGHT P(X>x)
```
=1-NORM.DIST(x,mu,sigma,TRUE)
```

Normal BETWEEN P(a<X<b)
```
=NORM.DIST(b,mu,sigma,TRUE)-NORM.DIST(a,mu,sigma,TRUE)
```

Inverse normal: area -> X
```
=NORM.INV(prob,mu,sigma)
```

Standard normal LEFT
```
=NORM.S.DIST(z,TRUE)
```

Inverse standard normal: area -> Z
```
=NORM.S.INV(prob)
```

Standardise X -> Z
```
=STANDARDIZE(x,mu,sigma)
```

NORMALITY / NORMAL PROBABILITY PLOT

Mean =AVERAGE(range) | Median =MEDIAN(range) | SD =STDEV.S(range)

Empirical share within 1 SD: =COUNTIFS(range,">="&(mean-sd),range,"<="&(mean+sd))/n

Plotting position
```
=j/(n+1)
```

Theoretical Z
```
=NORM.S.INV(alpha_cell)
```

Sort X ascending; plot Z on x-axis and ordered X on y-axis.

UNIFORM

Density height
```
=1/(b-a)
```

P(lower<X<upper)
```
=(upper-lower)/(b-a)
```

For endpoints inside support [a,b].

Mean
```
=(a+b)/2
```

Standard deviation
```
=(b-a)/SQRT(12)
```

EXPONENTIAL

LEFT P(X<t)
```
=EXPON.DIST(t,lambda,TRUE)
```

RIGHT P(X>t)
```
=1-EXPON.DIST(t,lambda,TRUE)
```

Percentile / inverse
```
= -LN(1-q)/lambda
```

lambda and time must use matching units.

BINOMIAL -> NORMAL APPROXIMATION

Use only when n*pi>5 AND n*(1-pi)>5.

Approx mean
```
=n*pi
```

Approx SD
```
=SQRT(n*pi*(1-pi))
```

Continuity correction: <= / > use k+0.5 ; < / >= use k-0.5.

Approx P(X<=k)
```
=NORM.DIST(k+0.5,n*pi,SQRT(n*pi*(1-pi)),TRUE)
```

Approx P(X>k)
```
=1-NORM.DIST(k+0.5,n*pi,SQRT(n*pi*(1-pi)),TRUE)
```

CHAPTER 7 - SAMPLING DISTRIBUTIONS

SAMPLE MEAN - POPULATION sigma KNOWN

Standard error
```
=sigma/SQRT(n)
```

P(Xbar < x0)
```
=NORM.DIST(x0,mu,sigma/SQRT(n),TRUE)
```

P(Xbar > x0)
```
=1-NORM.DIST(x0,mu,sigma/SQRT(n),TRUE)
```

P(a < Xbar < b)
```
=NORM.DIST(b,mu,sigma/SQRT(n),TRUE)-NORM.DIST(a,mu,sigma/SQRT(n),TRUE)
```

Inverse Xbar boundary
```
=NORM.INV(prob,mu,sigma/SQRT(n))
```

SAMPLE PROPORTION

Observed sample proportion
```
=successes/n
```

SE when population pi known
```
=SQRT(pi*(1-pi)/n)
```

*Current master practice rule: n*pi>5 AND n*(1-pi)>5.*

P(p < p0)
```
=NORM.DIST(p0,pi,SQRT(pi*(1-pi)/n),TRUE)
```

P(p > p0)
```
=1-NORM.DIST(p0,pi,SQRT(pi*(1-pi)/n),TRUE)
```

DIFFERENCE OF TWO INDEPENDENT MEANS

SE difference
```
=SQRT(sigma1^2/n1+sigma2^2/n2)
```

LEFT P(diff<c)
```
=NORM.DIST(c,mu1-mu2,SEdiff,TRUE)
```

RIGHT P(diff>c)
```
=1-NORM.DIST(c,mu1-mu2,SEdiff,TRUE)
```

FINITE POPULATION CORRECTION

FPC factor
```
=SQRT((N-n)/(N-1))
```

Adjusted standard error
```
=ordinary_SE*SQRT((N-n)/(N-1))
```

Without replacement; use course trigger n/N>0.05.

UNKNOWN sigma / STUDENT t ROUTE

t statistic
```
= (xbar-mu)/(s/SQRT(n))
```

Degrees of freedom
```
=n-1
```

t cumulative LEFT
```
=T.DIST(tcalc,df,TRUE)
```

t RIGHT
```
=1-T.DIST(tcalc,df,TRUE)
```

t BETWEEN = T.DIST(upper_t,df,TRUE) - T.DIST(lower_t,df,TRUE)

t boundary from LEFT area
```
=T.INV(prob,df)
```

CHAPTER 8 - CONFIDENCE INTERVALS

alpha
```
=1-C
```

Two-sided Z critical
```
=NORM.S.INV(1-alpha/2)
```

Two-sided t critical
```
=T.INV.2T(alpha,n-1)
```

MEAN CI

Z CI lower (sigma known)
```
=xbar-NORM.S.INV(1-alpha/2)*sigma/SQRT(n)
```

Z CI upper (sigma known)
```
=xbar+NORM.S.INV(1-alpha/2)*sigma/SQRT(n)
```

t CI lower (sigma unknown)
```
=xbar-T.INV.2T(alpha,n-1)*s/SQRT(n)
```

t CI upper (sigma unknown)
```
=xbar+T.INV.2T(alpha,n-1)*s/SQRT(n)
```

PROPORTION CI

p
```
=successes/n
```

Estimated SE
```
=SQRT(p*(1-p)/n)
```

*Check n*p>5 AND n*(1-p)>5 before the normal-approximation CI.*

CI lower
```
=p-NORM.S.INV(1-alpha/2)*SQRT(p*(1-p)/n)
```

CI upper
```
=p+NORM.S.INV(1-alpha/2)*SQRT(p*(1-p)/n)
```

REQUIRED SAMPLE SIZE

Mean
```
=ROUNDUP((NORM.S.INV(1-alpha/2)^2*sigma^2)/e^2,0)
```

Proportion
```
=ROUNDUP((NORM.S.INV(1-alpha/2)^2*pi*(1-pi))/e^2,0)
```

No prior pi estimate -> use pi=0.5.

FINITE POPULATION

CI adjusted SE
```
=ordinary_SE*SQRT((N-n)/(N-1))
```

Adjusted finite n
```
=(n0*N)/(n0+(N-1))
```

Use UNROUNDED ordinary n0 first.

Final finite required n
```
=ROUNDUP(adjusted_n,0)
```

BOOTSTRAP

Resample one value
```
=INDEX(data,RANDBETWEEN(1,COUNT(data)))
```

data = vertical numeric observed-sample range; WITH replacement.

Bootstrap mean
```
=AVERAGE(resample_range)
```

Lower ordered cutoff
```
=SMALL(stat_range,k)
```

Upper ordered cutoff
```
=LARGE(stat_range,k)
```

Use the course/question-specified cutoff position k.

CHAPTER 9 - HYPOTHESIS TESTING

Formal scope: Slides 1-51 / Lessons 9.1-9.7.

TEST STATISTICS

Z test (sigma known)
```
= (xbar-mu0)/(sigma/SQRT(n))
```

t test (sigma unknown)
```
= (xbar-mu0)/(s/SQRT(n))
```

df
```
=n-1
```

CRITICAL VALUES

Z lower one-tail
```
=NORM.S.INV(alpha)
```

Z upper one-tail
```
=NORM.S.INV(1-alpha)
```

Z two-tail magnitude
```
=NORM.S.INV(1-alpha/2)
```

t lower one-tail
```
=T.INV(alpha,df)
```

t upper one-tail
```
=T.INV(1-alpha,df)
```

t two-tail magnitude
```
=T.INV.2T(alpha,df)
```

P-VALUES

Z lower-tail
```
=NORM.S.DIST(zcalc,TRUE)
```

Z upper-tail
```
=1-NORM.S.DIST(zcalc,TRUE)
```

Z two-tail
```
=(1-NORM.S.DIST(ABS(zcalc),TRUE))*2
```

t lower-tail
```
=T.DIST(tcalc,df,TRUE)
```

t upper-tail
```
=1-T.DIST(tcalc,df,TRUE)
```

t two-tail
```
=T.DIST.2T(ABS(tcalc),df)
```

LESSON 9.6 - Z TEST <-> SAME-LEVEL Z CI

Matching Z CI lower
```
=xbar-NORM.S.INV(1-alpha/2)*sigma/SQRT(n)
```

Matching Z CI upper
```
=xbar+NORM.S.INV(1-alpha/2)*sigma/SQRT(n)
```

POWER

Basic relation
```
=1-beta
```

OFFICIAL PRACTICE Q4 BRIDGE ONLY - upper-tail known-sigma case.

Q4 critical Xbar
```
=mu0+NORM.S.INV(1-alpha)*sigma/SQRT(n)
```

Q4 power at true mu1
```
=1-NORM.DIST(critical_xbar,mu1,sigma/SQRT(n),TRUE)
```

Q4 beta
```
=1-power
```

AUDIT BASIS: supplied SDS188 final/frozen A1S2 system, Ch6/T6 normality + distribution resources, Ch7 formula sheet, Ch8/T8 finite-population + bootstrap resources, and Ch9 formal scope/practice-bridge. Q4 bridge: cumulative = LEFT area | required n = ROUND UP | FPC only without replacement | Ch9 broader post-9.7 power material excluded.